Beyond prescribed saturation: exact realization and numerical boundary geometry

24 September 2026. Current first-line research entry. This note connects four new proof notes to the retained integral obstruction theory; it does not replace their proofs or the earlier three-line history.

1. The main change in the mathematics

The preceding KE theorem used gauge saturation to force a half-flat bubble, then topology to exclude it. That is no longer the strongest route to **general Einstein existence**. A stronger existing ALE nonembedding result excludes every bubble under vanishing second homotopy. Saturation remains useful for identifying Kähler or locally symmetric structure, but is not necessary for existence in the bounded, noncollapsed regime.

The distinction is mathematical, not terminological:

general existence needs compactness and preservation of smooth type;

identifying a KE or symmetric output uses additional actual structure.

A KE metric already supplies the existential Einstein output. It must not be described as an output shortfall.

2. Pointwise-only realization for both signs

The all-ALE proof establishes

$$\boxed{\begin{array}{l}\pi_2(M)=0,\qquad \operatorname{diam}(g)\leqslant D,\qquad V_g\geqslant v>0,\\ \qquad\qquad \|\operatorname{Ric}_g^\sharp-\sigma I\|_\infty<\varepsilon(D,v)\end{array}} \implies \exists h \text{ on the same smooth } M:\operatorname{Ric}_h=\sigma h. \tag{1}$$

Here $\sigma=\pm1$; the threshold is uniform across the stated smooth types. There is no Weyl, saturation, monopole, or torsion-free fundamental-group hypothesis in (1).

The strengthened topological input is **classical**: Zhang's ALE nonembedding argument and Bamler–Chen, Lemma 4.35. Lift a proposed core to the universal cover of M. It may not lift as itself when its fundamental-group image is nontrivial, but it lifts as a finite cover, because an ALE Ricci-flat core has finite fundamental group. This lifted core is still nonflat Ricci-flat ALE, whereas its ambient has integer $H_1=H_2=0$. The classical lemma excludes it.

The previous mixed-chirality rational-ball alternative is therefore resolved in this topology. No classification of general Ricci-flat ALE spaces is needed. The detailed note explains the boundary-homology and eta mechanism, including why the quaternion example survives a group-order test but fails the curvature identities.

For $\sigma=+1$, Myers supplies a uniform diameter bound when $\delta_+<1/2$. The threshold in (1) then depends only on v. For $\sigma=-1$, an actual monopole projection $p\geqslant A_0>0$ gives

$$V_g\geqslant\frac{2\pi^2A_0}{(1+\delta_-)^2}.$$

It can replace the volume hypothesis, without any small gauge deficit. A positive-square integral actual class permits $A_0 = 1$.

This is an enlargement of the project's covered input regime. The proof credits the existing no-bubble and compactness theory; it does not establish global priority for the resulting corollary.

3. Two topology-free routes identify the exact geometry

Bilateral actual gauge input

The bilateral theorem uses actual conformal monopole structures in both orientations. With bounded diameter, one uniform positive projection floor, and

$$\delta_- + \frac{\mathfrak{D}_+}{V} + \frac{\mathfrak{D}_-}{V} \longrightarrow 0,$$

its uniform-threshold conclusion is an exact metric on the same smooth manifold whose universal cover is

$$\mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}.$$

No homotopy hypothesis remains. Both Weyl halves vanish on a hypothetical harmonic-radius blowup, making it flat before an ALE theorem is required. Two global actual forms pass to parallel opposite-orientation forms and give a parallel two-plane splitting. The quotient need not be a globally marked direct product.

The paper also replaces the two functional assumptions by a single raw Ricci-energy excess under actual projection budgets. This is an exact identity, not an additional independent discovery. Existing non-Kähler shear examples satisfy bilateral saturation, so deleting diameter from this KE/bidisk statement is actually false.

One actual class at the Miyaoka–Yau numerical boundary

The Chern-boundary theorem assumes

$$A = 2\chi + 3\tau = 3(2\chi - 3\tau) > 0, \qquad p \geqslant A.$$

It has **no separately imposed saturation condition**. Classical identities give, with $P = \int |\mathrm{Ric}^\circ|^2$,

$$S - 32\pi^2 A + 48W_- = 12P, \qquad \mathfrak{D}_2 + 72\pi^2(p - A) + 54W_- = 18P.$$

Consequently

$$\frac{\mathfrak{D}_2}{V} \leqslant 72\delta_-^2, \qquad \frac{1}{V}\int |W^-|^2 \leqslant \delta_-^2.$$

Thus small pointwise defect and bounded diameter imply complex-hyperbolic geometry on the same oriented smooth manifold. Again there is no π_2 or fundamental-group assumption. The classical exact Miyaoka–Yau classification is not being rediscovered: this is its explicitly proved approximate-to-exact composition in the bounded-diameter setting.

4. A different realization route does not need diameter

The local-to-global proof first shows

$$\pi_2 = 0 + \text{bounded Ricci} + \inf_x \mathrm{Vol}\, B_1(x) > 0 \implies \text{uniform local harmonic radius.} \tag{2}$$

Only one local Jiang–Naber energy budget is needed for a blowup, so total volume and diameter do not enter.

A classical Bianchi-gauge solver then produces an exact negative Einstein metric if there is also a full tensor L^2 spectral gap, bounded $\nabla \mathrm{Ric}$, and a sufficiently small mixed residual

$$\delta_-^{1-\alpha} + \| \mathrm{Ric} + g \|_{L^2}.$$

The proof specifies the inverse, mixed Hölder–Sobolev norms, nonlinear estimates, positivity radius, and gauge removal. It does not obtain the global inverse from local regularity. Teichmüller kernels are not deleted, and the L^2 residual is unnormalized. Bounded-volume support of the residual gives a direct pointwise-smallness corollary even when the surrounding exact region has unbounded volume.

This is a useful sufficient criterion with a classical solver, not a claim to have solved arbitrary large-diameter degeneration.

5. What this changes for Yau26

Within $\pi_2 = 0$, a sequence of manifolds with **no Einstein metric** and pointwise defect tending to zero must obey:

- Positive sign: total volume tends to zero, since Myers already bounds diameter and (1) rules out every positive-volume subsequence.
- Negative sign with a uniform positive actual projection budget: diameter tends to infinity, since the budget prevents total-volume collapse.

These are deductions about possible original-problem witnesses, not constructions of such witnesses. Nonzero π_2, uncontrolled collapse, and large-scale global inversion remain outside (1). The fixed non-Kähler shear objects still have undecided general Einstein existence.

The first-line structure is now clearer: the integral theory supplies diameter-free obstructions in incompatible topological regions; the compact regime supplies general exact realization; actual saturation and numerical equality identify stronger output geometry; global spectral control gives a separate noncompactness-tolerant repair route. There is no reason to force all four jobs into one auxiliary-form estimate.

6. Proof and artifact status

All four statements have proofs in the linked English notes, with classical inputs and inherited PDE dependencies identified. Finite checks verify the displayed identities, real model calibrations, and normalization only. They do not independently certify the geometric proofs or originality.

Earlier proofs, source archives, live Sessions, and installed Skills are unchanged. The previous realization and rigidity entry remains the account of the preceding stage. Current catalogue IDs are I24–I27; these identify result families, not four independent claims of world-first discovery.